

Foundation (Jalapeno)

- Q1.** Solve the system using substitution: $x + 2y = 4$ and $x - 2y = -2$. What is y ?
- (A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$
- Q2.** Solve the system using elimination: $3x + 2y = 7$ and $2x - 2y = -3$. What is x ?
- (A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$
- Q3.** A spaceship is traveling through space. Its x and y coordinates are given by the system: $x - 2y = 5$ and $3x + 2y = 11$. What is x ?
- (A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$
- Q4.** A scientists is conducting an experiment with two variables, x and y . The system of equations is: $2x + 3y = 12$ and $x - 2y = -3$. What is y ?
- (A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$
- Q5.** Solve the system using substitution: $x + y = 4$ and $x - y = 2$. What is x ?
- (A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$
- Q6.** A planet's orbit can be modeled by the system: $2x - 3y = 5$ and $x + 2y = 7$. What is y ?
- (A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$
- Q7.** Solve the system using elimination: $x + 2y = 5$ and $3x - 2y = 7$. What is x ?
- (A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$
- Q8.** A lab experiment involves two variables, x and y . The system of equations is: $x - y = 2$ and $2x + 3y = 12$. What is x ?
- (A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$

Open Ended Questions (FRQ)

- Q9.** Solve the system using substitution or elimination: $2x + 5y = 17$ and $x - 2y = -3$. Show all work and find the values of x and y .
- Q10.** A spaceship's trajectory can be modeled by the system: $3x - 2y = 5$ and $2x + 5y = 23$. Determine the number of solutions to the system and explain your reasoning.

Chef's Tips

- Review the concept of substitution and elimination methods
- Emphasize the importance of checking the solution
- Use real-world examples to illustrate the application of systems of linear equations